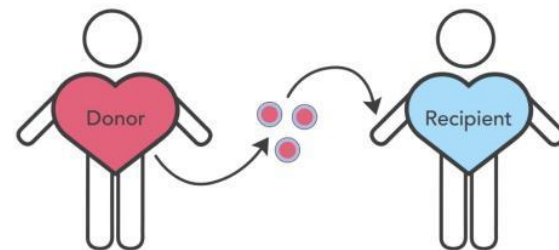




Post-HCT Surgery Survival: Analysis and Prediction

Group 521:

- Behafarin Emam
- Likhita Nallapati
- Shriya Rawal
- Jackson Aghazadeh



****slide to remove**** Powerpoint Topics we need to include

Overview of the project

Discuss your team (individual skills and contributions in the group project)

Identify and describe the data you used

Discuss your analysis of stakeholders, use, and intentions

Discuss your analytic approach

Describe your findings / results

Overview

Objective

Dataset Analysis

Uses and Intention

Analytic Approach + Findings

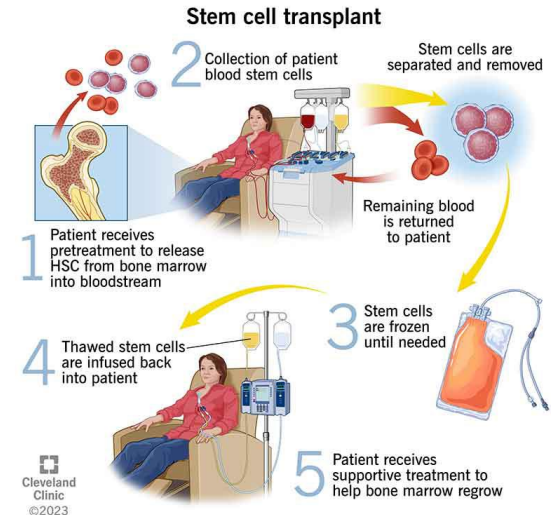
Limitations and Challenges

Objective

- Inspired by [Kaggle Competition](#) on Post-Hematopoietic Cell Transplantation (HCT) survival predictions
- The aim is to analyze survival outcomes for patients undergoing HCT

What is Hematopoietic Cell Transplantation?

- HCT is a medical procedure that replaces damaged or diseased blood-forming cells with healthy ones
- This procedure is vital for treating hematological diseases such as leukemia, lymphoma, and certain blood disorders.



Performance Metric

Concordance index

The concordance index is a value between 0 and 1 where:

- 0.5 is the expected result from random predictions,
- 1.0 is a perfect concordance and,
- 0.0 is perfect anti-concordance (multiply predictions with -1 to get 1.0)

$$\text{C-index} = \frac{\sum_{i,j} 1_{T_j < T_i} \cdot 1_{\eta_j > \eta_i} \cdot \delta_j}{\sum_{i,j} 1_{T_j < T_i} \cdot \delta_j}$$

with:

- η_i , the risk score of a unit i
- $1_{T_j < T_i} = 1$ if $T_j < T_i$ else 0
- $1_{\eta_j > \eta_i} = 1$ if $\eta_j > \eta_i$ else 0

Dataset Analysis

This dataset is suited for both survival prediction and equity-focused analyses.

Demographics

- ethnicity and race group

Treatment details

- HLA match, graft type, and conditioning intensity

28,800 rows
60 columns

Medical History

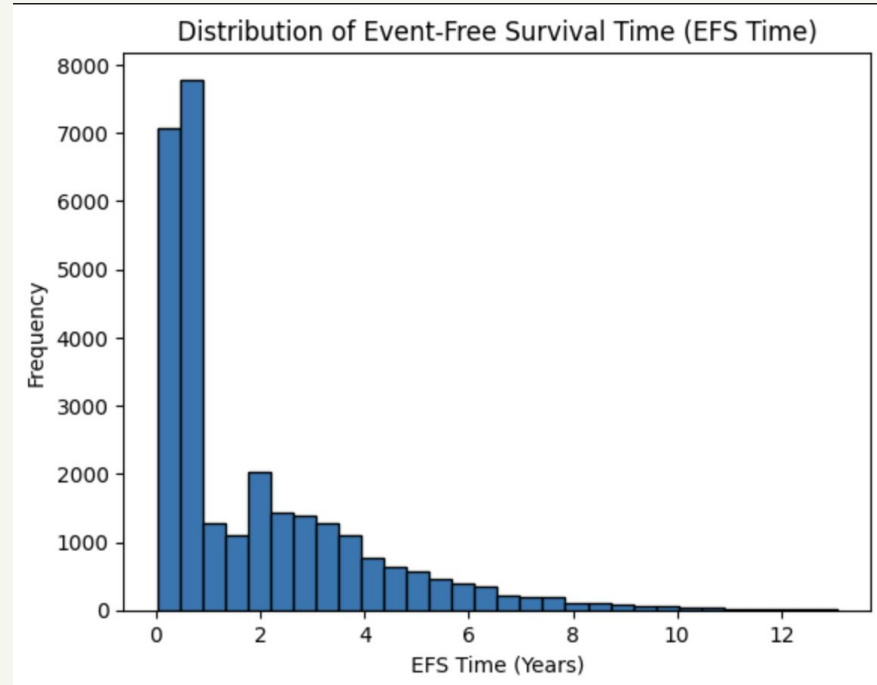
- comorbidity scores, prior tumors, pre-existing conditions: diabetes and cardiac issues

Outcome variables

- Event-Free Survival & EFS time representing survival status and time to an event or censoring.

Distribution of Event-Free Survival Time

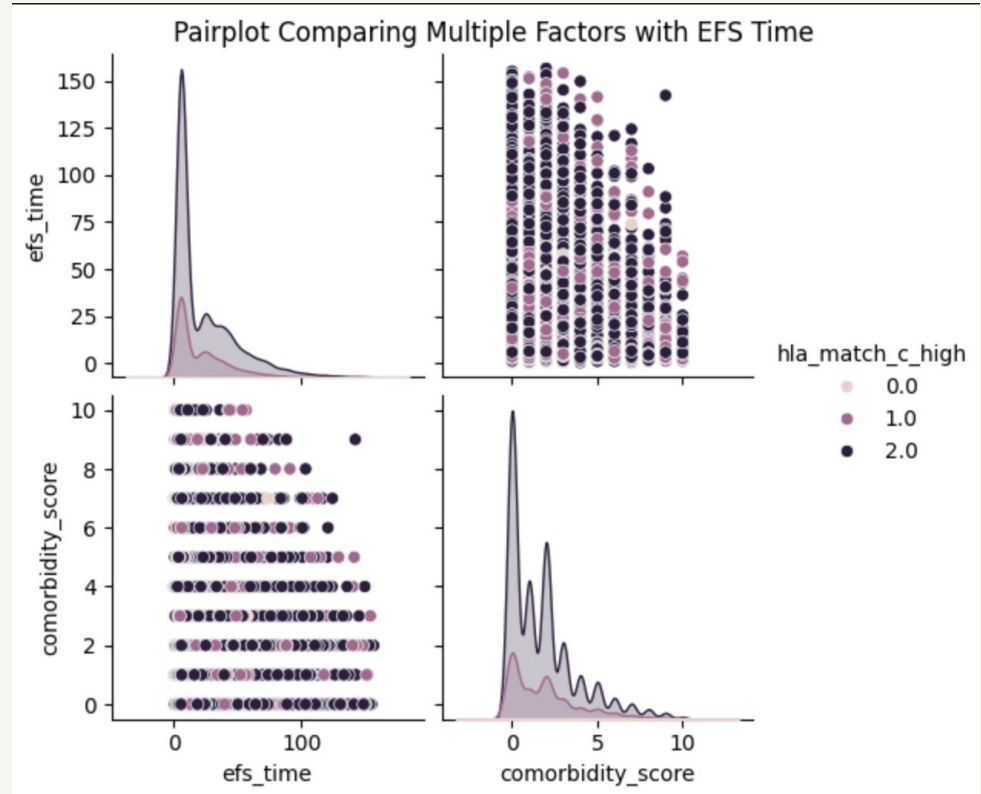
- The data is right-skewed, with most patients experiencing events within the first two years post-HCT.
- The events here refer to any post-op complication that they may have suffered.



Pairplot

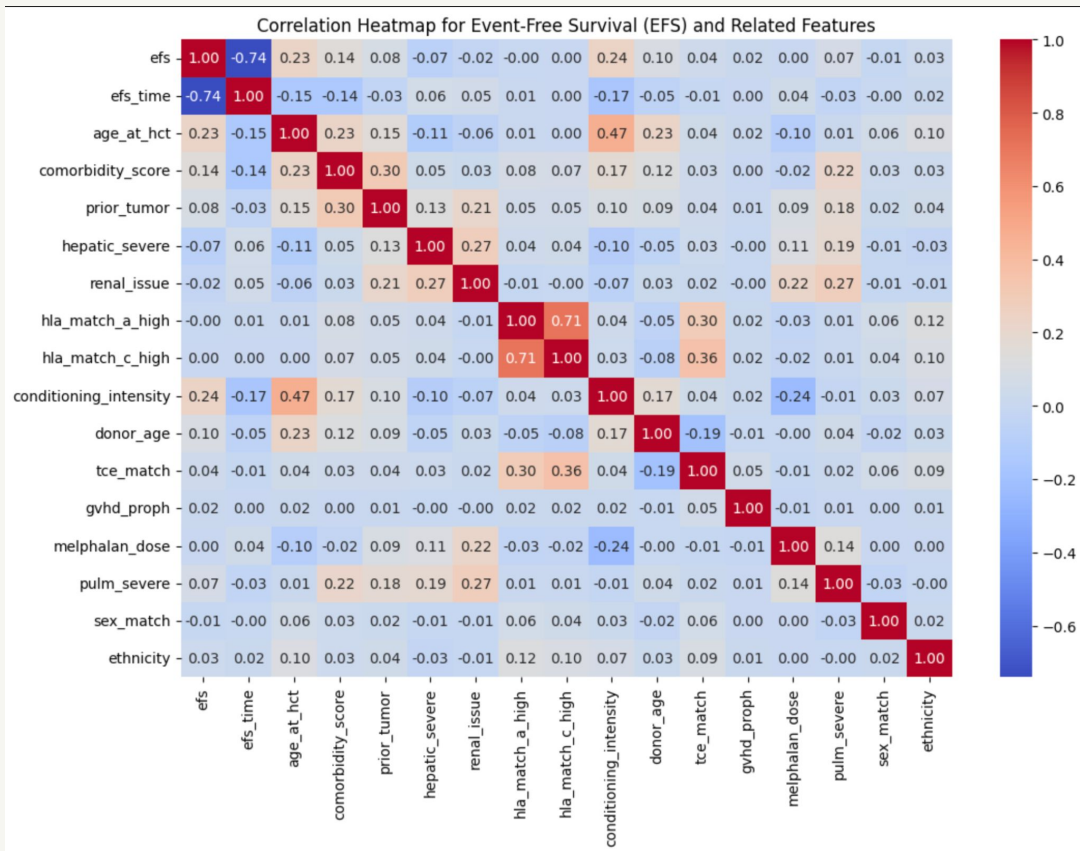
The pairplot examines relationships between EFS time, comorbidity scores, and HLA matching (hla_match_c_high)

- Show patterns in survival times based on these key factors: suggests that higher comorbidity scores may correspond to shorter EFS times.



Correlation Heatmap

-Key correlations include the relationship between conditioning intensity and age, as well as associations between HLA matching and survival outcomes, which provide insights into significant predictors.

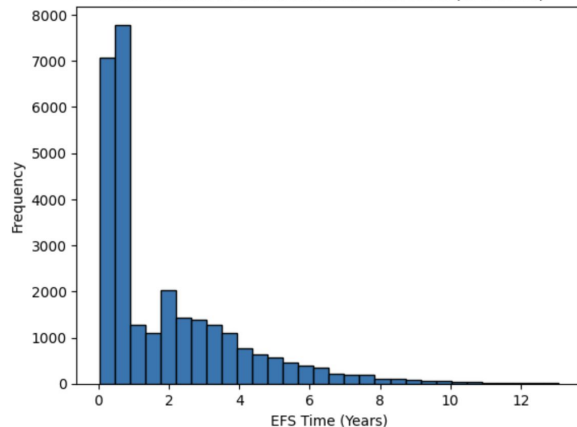


Event-Free Survival by Race Group

Final Summary with Percentage of efs and Average efs_time:

	race_group	0.0	1.0	avg_efs_time
0	American Indian or Alaska Native	48.96	51.04	24.16
1	Asian	44.85	55.15	22.63
2	Black or African-American	44.13	55.87	21.81
3	More than one race	53.42	46.58	25.91
4	Native Hawaiian or other Pacific Islander	47.65	52.35	23.07
5	White	37.45	62.55	21.83

Distribution of Event-Free Survival Time (EFS Time)

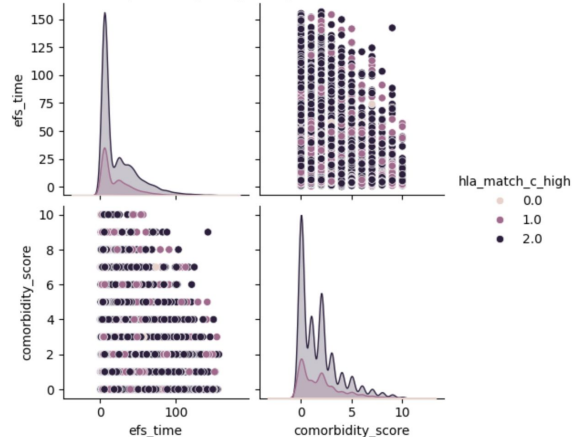


This histogram displays the distribution of event-free survival time (EFS Time) in years across all patients in the dataset.

- The data is right-skewed, with most patients experiencing events within the first two years post-HCT.

- The events here refer to any post-op complication that they may have suffered.

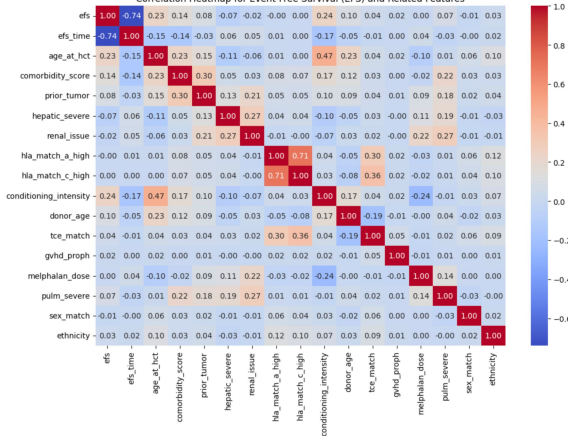
Pairplot Comparing Multiple Factors with EFS Time



The pairplot examines relationships between EFS time, comorbidity scores, and HLA matching (hla_match_c_high)

- Show patterns in survival times based on these key factors: suggests that higher comorbidity scores may correspond to shorter EFS times.

Correlation Heatmap for Event-Free Survival (EFS) and Related Features



The heatmap highlights correlations between various features and event-free survival outcomes.

- Key correlations include the relationship between conditioning intensity and age, as well as associations between HLA matching and survival outcomes, which provide insights into significant predictors.

Summarize Finding? Or add another graph??

Final Summary with Percentage of efs and Average efs_time:

	race_group	0.0	1.0	avg_efs_time
0	American Indian or Alaska Native	48.96	51.04	24.16
1	Asian	44.85	55.15	22.63
2	Black or African-American	44.13	55.87	21.81
3	More than one race	53.42	46.58	25.91
4	Native Hawaiian or other Pacific Islander	47.65	52.35	23.07
5	White	37.45	62.55	21.83

Stakeholders and Applications



1

Healthcare Providers

Hematologists, Oncologists, and Transplant specialists

It will assist in making more informed clinical decisions, tailoring treatments to individual patients, and improving patient care based on the model's insights into survival probabilities.

2

Patients and Patient Advocacy Groups

Specifically those focusing on blood cancers, bone marrow diseases, and stem cell transplants

Provides potential predictors of survival and could help patients better understand their treatment options and prepare for the challenges associated with HCT.

3

Academic Institutions

Institutions and educators in fields of medicine, data science, bioinformatics, public health, and health policy

Would be useful for a case study or teaching tool, to help illustrate the practical applications of predictive analytics and the importance of equity-focused research in healthcare.

4

Medical Researchers

Medical researchers focusing on hematology, oncology, transplant medicine, and health equity

Used for predictive model and its data analysis to uncover new insights or to inform future studies, specifically for survival disparities related to socioeconomic and demographic factors.

5

Bioinformaticians and Analysts

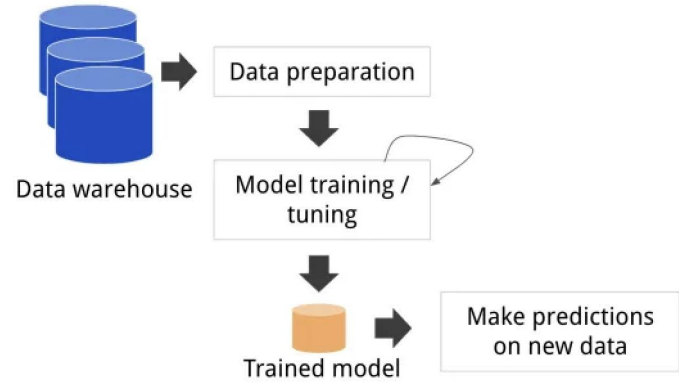
Data scientists, bioinformaticians, and healthcare analysts involved in predictive modeling and data-driven healthcare insights

Create further methods used to develop the model and how it addresses health equity, in terms of fairness and bias in predictive models.

6

**** ADD one more****

Analytic Approach



Data Preprocessing

Model Creation

Prediction and Analysis

Data Preprocessing and Feature Handling

Total of 60 Features

35 Categorical Features , 25 Numerical Features

- Handling categorical & missing values
- One-hot encoding & ordinal encoding
- Removing highly correlated features
- Standardization of numerical features

```
diabetes
```

```
[8] train_df[train_df['diabetes'].isnull()][['ID','efs']].groupby('efs').count()
```

efs	ID
0.0	952
1.0	1167

```
train_df['diabetes'] = train_df['diabetes'].fillna('Unknown')  
diabetes_mapping = {'No': 0, 'Yes': 1, 'Not done': 2, 'Unknown': 3}  
train_df['diabetes'] = train_df['diabetes'].map(diabetes_mapping)
```

```
# Encoding the dri_score feature  
dri_score_mapping = {  
    'Low': 1,  
    'Intermediate': 2,  
    'Intermediate - TED AML case <missing cytogenetics': 2.5,  
    'High': 3,  
    'High - TED AML case <missing cytogenetics': 3.5,  
    'Very high': 4,  
    'N/A - disease not classifiable': -1,  
    'N/A - non-malignant indication': -2,  
    'N/A - pediatric': -2,  
    'TBD cytogenetics': -1,  
    'Missing disease status': -1  
}  
  
# Apply the mapping  
train_df['dri_score'] = train_df['dri_score'].map(dri_score_mapping)  
median_dri_score = train_df['dri_score'].median()  
train_df['dri_score'] = train_df['dri_score'].fillna(median_dri_score)
```


Models used: Kaplan Meier Curve + Xgboost + _ _ _ _

Prediction and Analysis

Prediction results

Feature importance extracted from Xgboost model:

Challenges and Limitations